

Newton forward interpolation

$x: x_0, x_1, x_2, x_3, \dots, x_n$
 $y: y_0, y_1, y_2, y_3, \dots, y_n$

$h = x_1 - x_0$

$$y(x_0+h) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

Where $p = \frac{x - x_0}{h}$

$$y(x_0+h) = y_0 + p \Delta y_0 + \frac{p^2 - p}{2} \Delta^2 y_0 + \frac{p^3 - 3p^2 + 2p}{6} \Delta^3 y_0 + \frac{p^4 - 6p^3 + 11p^2 - 6p}{24} \Delta^4 y_0 + \dots$$

$$\frac{p^5 - 10p^4 + 35p^3 - 50p^2 + 24p}{120} \Delta^5 y_0 + \dots$$

$$y''(x_0+h) = \frac{1}{h^2} \left[\Delta^2 y_0 + \frac{2p-1}{2} \Delta^3 y_0 + \frac{3p^2 - 6p + 2}{6} \Delta^4 y_0 + \frac{4p^3 - 18p^2 + 22p - 6}{24} \Delta^5 y_0 + \dots \right]$$

$$\frac{5p^4 - 40p^3 + 105p^2 - 100p + 24}{120} \Delta^5 y_0 + \dots$$

$$y'''(x_0+h) = \frac{1}{h^3} \left[\Delta^3 y_0 + \frac{(p-1)}{2} \Delta^4 y_0 + \frac{12p^2 - 36p + 22}{24} \Delta^5 y_0 + \frac{20p^3 - 120p^2 + 210p - 100}{120} \Delta^6 y_0 + \dots \right]$$

Note: at $x = x_0$ then $p = \frac{x - x_0}{h} = 0$

then $y' = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 + \dots \right]$

$y'' = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right]$

From the following data find the value of $\frac{dy}{dx}$ ~~at~~ $\frac{d^2y}{dx^2}$ 80

At the point $x=1.5$

x : 1.5 2 2.5 3.0 3.5 4.0

y : 3.375 7 13.625 24 38.875 59

$$y' = y_0 + p\Delta y_0 + p^2\Delta^2 y_0$$

$$h = x_1 - x_0 = 2 - 1.5 = 0.5$$

$$x = x_0$$

$$y' = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right] \rightarrow \textcircled{1}$$

Forward diff table

x	y	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$
1.5	3.375			
2	7	3.625		
2.5	13.625	6.625	3	
3.0	24	10.375	3.75	0.75
3.5	38.875	14.875	4.5	0.75
4	59	20.125	5.25	0.75

$$\textcircled{1} \rightarrow = \frac{1}{0.5} \left[3.625 - \frac{1}{2} (3) + \frac{1}{3} (0.75) - \frac{1}{4} (0) + \frac{1}{5} (0) \right]$$

$$= \underline{2.875} \times \frac{2.375}{0.5} = 4.75$$

$$y'' = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$= \frac{1}{0.25} \left[3 - 0.75 + \frac{11}{12} (0) - \frac{5}{6} (0) \right]$$

$$= \underline{2.25} \times \frac{2.25}{0.25} = 9$$

From the following data find $\frac{dy}{dx}$ & $\frac{d^2y}{dx^2}$ at the point 81

$$x=1 \text{ \& } x=1.1$$

x:	1	1.2	1.4	1.6	1.8	2.0
y:	0	0.128	0.544	1.296	2.432	4

Forward diff table

x	y	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$
1	0			
1.2	0.128	0.128	0.288	0.048
1.4	0.544	0.416	0.336	0.048
1.6	1.296	0.752	0.384	0.048
1.8	2.432	1.136	0.432	
2.0	4	1.568		

$$y' = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$= \frac{1}{0.2} \left[0.128 - \frac{1}{2} 0.288 + \frac{1}{3} (0.048) + 0 \right]$$

$$= \underline{\underline{0}}$$

$$y'' = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$= \frac{1}{0.04} \left[0.288 - 0.048 \right] = \underline{\underline{6}}$$

When $x=1.1$ $h = x_1 - x_0 = 1.2 - 1 = 0.2$ $p = \frac{x - x_0}{h} = \frac{1.1 - 1}{0.2} = 0.5$

$$y' = \frac{1}{h} \left[\Delta y_0 + \frac{2p-1}{2} \Delta^2 y_0 + \frac{3p^2-6p+2}{6} \Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{24} \Delta^4 y_0 + \frac{5p^4-40p^3+105p^2-100p+24}{120} \Delta^5 y_0 \right]$$

$$= \frac{1}{0.2} \left[0.128 + \frac{2(0.5)-1}{2} 0.288 + \frac{3(0.5)^2 - 6(0.5) + 2}{6} (0.048) + 0 \right]$$

$$= \frac{1}{0.2} \left[\right] = \underline{\underline{0.63}}$$

$$y''(1.1) = \frac{1}{h^2} \left[\Delta^2 y_0 + (p-1) \Delta^3 y_0 \right]$$

$$= \frac{1}{(0.2)^2} \left[0.288 + (0.5-1) 0.048 \right] = \underline{\underline{6.6}}$$

Newton backward interpolation:-

$$x: x_0 \quad x_1 \quad x_2 \quad \dots \quad x_n$$

$$y: y_0 \quad y_1 \quad y_2 \quad \dots \quad y_n$$

$$y(x_0+h) \text{ at } x=x_n$$

$$y(x_0+h) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n$$

$$\text{where } p = \frac{x - x_0}{h}$$

$$y(x_0+h) = y_n + p \nabla y_n + \frac{p^2+p}{2} \nabla^2 y_n + \frac{p^3+3p^2+2p}{6} \nabla^3 y_n + \frac{p^4+6p^3+11p^2+6p}{24} \nabla^4 y_n$$

$$y'(x_0+h) = \frac{1}{h} \left[0 + \nabla y_n + \frac{2p+1}{2} \nabla^2 y_n + \frac{3p^2+6p+2}{6} \nabla^3 y_n + \frac{4p^3+18p^2+22p+6}{24} \nabla^4 y_n \right]$$

$$y'' = \frac{1}{h^2} \left[\cancel{\Delta^0 y_n} + \nabla^2 y_n + 6p + 6(p+1) \nabla^3 y_n + \frac{12p^2+36p+22}{24} \nabla^4 y_n \right]$$

$$y'' = \frac{1}{h^2} \left[\nabla^2 y_n + (p+1) \nabla^3 y_n + \frac{6p^2+18p+11}{12} \nabla^4 y_n + \dots \right] \checkmark$$

if $x = x_n$ then $p=0$

$$y' = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right] \checkmark$$

$$y'' = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n \right] \checkmark$$

1) A particle is moving along a straight line the displacement x at some ~~inter~~ time t are given below. Find the velocity & acceleration of the particle at $t=4$.

t	0	1	2	3	4
x	5	8	12	17	26

$p=0$
 $h=1$

$$x' = \frac{1}{h} \left[\nabla x_n + \frac{2p+1}{2} \nabla^2 x_n + \frac{3p^2+6p+2}{6} \nabla^3 x_n + 4p^3 \right]$$

$$x' = \frac{1}{h} \left[\nabla x_n + \frac{1}{2} \nabla^2 x_n + \frac{1}{3} \nabla^3 x_n + \frac{1}{4} \nabla^4 x_n \right]$$

$$= \frac{1}{1} \left[9 + \frac{1}{2} (4) + \frac{1}{3} (3) + \frac{1}{4} (3) \right] = \underline{\underline{12.75}}$$

Forward table

t	x	∇x_n	$\nabla^2 x_n$	$\nabla^3 x_n$	$\nabla^4 x_n$
0	5	3	1	0	
1	8	4	1		
2	12	5			
3	17				
4	26	9	4	3	3

Newton's divided difference:

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Suppose the funⁿ $y=f(x)$ is known at $n+1$ unequally spaced points say $x: x_0, x_1, \dots, x_n$ then Newton's

divided diff interpolation is given by

$$y(x) = y_0 + (x-x_0)\Delta y_0 + (x-x_0)(x-x_1)\Delta^2 y_0 + (x-x_0)(x-x_1)(x-x_2)\Delta^3 y_0 + \dots$$

$$y'(x) = \Delta y_0 + \Delta^2 y_0 [(x-x_0) \cdot 1 + (x-x_1) \cdot 1] + \Delta^3 y_0 [(x-x_0)(x-x_1) \cdot 1 + (x-x_0)(x-x_2) \cdot 1 + (x-x_1)(x-x_2) \cdot 1] + \dots$$

1)

x	0	2	3	4	7	9	find $f'(5)$
y	4	26	58	112	466	922	

x	y	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$
0	4				
2	26	22			
3	58	32	10		
4	112	54	22	12	
7	466	354	118	16	4
9	922	228	22	6	0

$$y'(x) = \Delta y_0 + (x-x_0)\Delta y_0 + (x-x_0)\Delta^2 y_0 [(x-x_0) + (x-x_1)] + \Delta^3 y_0 [(x-x_0)(x-x_1) + (x-x_0)(x-x_2) + (x-x_1)(x-x_2)]$$

$$= 11 + 7[(5-0) + (5-2)] + 1[(5-0)(5-2) + (5-0)(5-3) + (5-2)(5-3)]$$

$$= 11 + 56 + 15 + 10 + 6 = 98$$

2) Find the 1st derivative of the funⁿ tabulated below at the point $x=10$.

x :	3	5	11	27	34
y :	-13	23	899	17315	35606

x	y	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$
3	-13				
5	23	18			
11	899	146	16		
27	17315	1026	40	1	
34	35606	2613	69	1	0

$$\begin{aligned}
 y'(x) &= \Delta y_0 + \Delta^2 y_0 [(x-x_0) + (x-x_1)] + \Delta^3 y_0 [(x-x_0)(x-x_1) + (x-x_0)(x-x_2) + (x-x_1)(x-x_2)] \\
 &= 18 + 16 [(10-3) + (10-5)] + 1 [(10-3)(10-5) + (10-3)(10-11) + (10-5)(10-11)] \\
 &= 18 + 16 [12] + [35 - 7 - 5] \\
 &= \underline{\underline{233}}
 \end{aligned}$$

3) find $f'(8)$

x :	4	5	7	10	11	13
y :	48	100	294	900	210	2028

x	y	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$
4	48			
5	100	52		
7	294	97	15	
10	900	202	21	1
11	210	310	27	1
13	2028	409	33	1

$$\begin{aligned}
 y'(x) &= \Delta y_0 + \Delta^2 y_0 [(x-x_0) + (x-x_1)] + \Delta^3 y_0 \left[\frac{(x-x_0)(x-x_1)(x-x_2)}{2} + (x-x_1)(x-x_2) \right] \\
 &= 52 + 15 [(8-4) + (8-5)] + 1 \left[\frac{(8-4)(8-5)(8-7)}{2} + (8-5)(8-7) \right] \\
 &= 52 + 15 [7] + [12 + 4 + 3] \\
 &= \underline{\underline{176}}
 \end{aligned}$$

Simpson's $\frac{1}{3}$ rule for Double integral:-

$$I = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots) + 2(y_2 + y_4 + y_6 + \dots) \right] \rightarrow \text{single integration}$$

Find the value of $\int_1^2 \int_1^2 \frac{dxdy}{x^2+y^2}$ using Simpson's $\frac{1}{3}$ rule taking

$h=k=0.25$

$x \rightarrow$		1	1.25	1.5	1.75	2	
$y \downarrow$							
1	$1/2.5$	0.39024	0.3077	0.2662	0.2	y_0	
1.25	0.39024	0.32	0.2623	0.2162	0.1798	y_1	
1.5	0.3077	0.2623	0.222	0.1882	0.16	y_2	
1.75	0.2662	0.2162	0.1882	0.1633	0.1416	y_3	
2	0.2	0.1798	0.1600	0.1416	0.125	y_4	

$$\begin{aligned}
 I &= \int_1^2 f(1,y) dy \\
 &= \frac{0.25}{3} \left[(0.5 + 0.2) + 4(0.39024 + 0.2462) + 2(0.30769) \right] \\
 &= \underline{\underline{0.3217}}
 \end{aligned}$$

$$I_1 = \int_1^2 f(1.25, y) dy$$

$$= \frac{0.25}{3} \left[(0.39024 + 0.17978) + 4(0.32 + 0.21622) + 2(0.26230) \right]$$

$$= \underline{\underline{0.27}}$$

$$I_2 = \int_1^2 f(1.5, y) dy$$

$$= \frac{0.25}{3} \left[(0.3077 + 0.16) + 4(0.2623 + 0.1882) + 2(0.222) \right]$$

$$= \underline{\underline{0.2261}}$$

$$I_3 = \int_1^2 f(1.75, y) dy$$

$$= \frac{0.25}{3} \left[(0.2462 + 0.1416) + 4(0.2162 + 0.1633) + 2(0.1882) \right]$$

$$= \underline{\underline{0.1902}}$$

$$I_4 = \int_1^2 f(2, y) dy$$

$$= \frac{0.25}{3} \left[(0.2 + 0.125) + 4(0.1798 + 0.1416) + 2(0.16) \right]$$

$$= \underline{\underline{0.1609}}$$

$$zI = \frac{k}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots) + 2(y_2 + y_4 + y_6 + \dots) \right] \quad 89$$

$$I = \frac{0.25}{3} \left[(I_0 + I_4) + 4(I_1 + I_3) + 2(I_2) \right]$$

$$= \frac{0.25}{3} \left[(0.3217 + 0.1609) + 4(0.27 + 0.1902) + 2(0.2261) \right]$$

$$= \underline{\underline{0.2313}}$$

$$2) \int_1^{1.5} \int_1^2 \frac{dx dy}{x+y}$$

$h = 0.5$ along x axis & $k = 0.25$ along y axis

$x \rightarrow$	1	1.5	2
$y \downarrow$			
1	0.5	0.4 0.3636	0.333
1.25	0.4444	0.3636	0.3077
1.5	0.4	0.333	0.2857

$$I_0 = \int_1^2 f(1, y) dy$$

$$= \frac{0.5}{3} \left[(0.5 + 0.4) + 4(0.3636) \right] = 0.3924 \quad 0.4463$$

$$I_1 = \int_1^2 f(1.5, y) dy$$

$$= \frac{0.5}{3} \left[(0.4 + 0.333) + 4(0.3636) \right] = 1.5766 \quad 0.3646$$

$$I_2 = \int_1^2 f(2, y) dy$$

$$= \frac{0.5}{3} \left[(0.333 + 0.2857) + 4(0.3077) \right] = 0.3083$$

$$I = \frac{k}{3} [(I_0 + I_3) + 4(I_1)]$$

$$= \frac{0.25}{3} [(0.4463 + 0.3083) + 4(0.3646)]$$

$$I = \underline{0.1844}$$